Deep Latent Spectral Representation Learning-Based Hyperspectral Band Selection for Target Detection

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***Abstract*— Hyperspectral images (HSIs) can provide discriminative spectral signatures regarding the physical nature of different materials. It is this unique nature that makes HSIs to be of great interest in many fields. However, HSI application faces various challenges due to high dimensionality, redundant information, noisy bands, and insufficient samples. To address these problems, we propose an unsupervised band selection method based on deep latent spectral representation learning, called DLSRL, in this article. It imposes spectral consistency on deep latent space that resolves the issue of insufficient samples and spectral information lost in HSI interpretation. It pursues the low-dimensional optimal representation of the high-dimensional HSIs. In particular, an adaptive mapping relationship is constructed between the deep latent representation and the optimal subset to preserve physical significance optimally. Furthermore, a hierarchical optimization approach is introduced to achieve target detection with the selected subset. To verify the superiority of the proposed method, experiments have been conducted on four data sets captured by different sensors over different scenes. Comparative analyses validate that the proposed method presents superior performance in terms of high detection accuracy and low false alarm rate.**

***Index Terms*— Band selection, hierarchical optimization, hyperspectral image (HSI), representation learning, spectral consistency.**

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I. Introduction

HYPERSPECTRAL imaging sensor can densely capture spectral information regarding the physical nature of the materials in an image scene. Benefitting from such rich and reliable spectral information, hyperspectral images (HSIs) have obvious superiority in military and civilian fields, compared with panchromatic and multispectral images [1]–[3]. However, the high dimensionality results in "Hughes phenomenon" as well as heavy computation burden, which restricts the successful application of HSIs [4]. Besides, strong correlations between adjacent bands results in a large amount of redundant information being processed, stored, and transmitted. To mitigate these problems, feature extraction and band selection are two generally applied techniques [5]–[7]. Feature extraction transforms a high-dimensional data into a lower dimensional space, while band selection pursues an optimal subset of hyperspectral bands and discards the unselected bands. Both these significant preprocessing steps can reduce redundancy or noisy information and thus have positive influence on HSI application to a certain degree. Different from feature extraction which transforms data, band selection can preserve the physical significance from original hyperspectral bands, forcing the reduced subset to be more comprehensible [8], [9].

In the last decade, a variety of band selection techniques have been developed, which can be roughly classified into three categories in accordance with the involvement of label prior information: 1) supervised [10]; 2) unsupervised [11]; and 3) semisupervised [12] methods. Supervised methods try to select band subset with the guidance of prior labels [13], [14] and semisupervised methods select subset by simultaneously employing the advantages of labeled and unlabeled samples [15], [16]. The limited label information restricts the real applications of supervised and semisupervised methods in most cases. Besides, supervised and semisupervised methods may lead to inappropriate selection

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due to the divergent spectral characteristics in the training samples among the same classes. In other words, generalization and stability of such methods are limited. On the contrary, unsupervised methods construct the selection rules depending on the given data itself without any label knowledge, which can be ranking-based, clustering-based, greedy-based, and evolutionary-based methods according to selection strategy [9], [17]. The ranking-based methods rank hyperspectral bands and then select the top-rank bands, such as principal component analysis (PCA) [18], maximum-variance PCA (MVPCA) [19], and linearly constrained minimum variance band correlation constraint (LCMV-BCC) [20]. The clustering-based methods search the optimal subset by first separating all the bands from groups and then selecting representative bands in each group such as enhanced density-peak-based clustering (E-FDPC) [21] and optimal clustering framework (OCF) [9]. The currently selected band of the greedy-based methods is based on the previous results in each iteration, such as volume gradient band selection (VGBS) [22]. The evolutionary-based methods first generate an initial subset and then update it to constitute an optimal subset such as particle swarm optimization (PSO) [23] and multitask sparsity pursuit (MTSP) [24]. In summary, the state-of-the-art methods may not necessarily represent the maximum information of all hyperspectral bands. Worse even, a few methods consider the underlying characteristics, such as multivariate Gaussianity and spectral consistency, for insufficient samples in HSIs.

In recent years, deep neural networks (DNNs) have been successfully applied in the field of HSIs, especially for feature extraction, enhancement, and classification. With respect to band selection, Zhan *et al.* [25] adopted a convolutional neural network (CNN) to achieve band selection with the help of training samples and conducted hyperspectral classification, which belongs to supervised methods. As mentioned above, supervised methods are limited due to their instability and weak generalization. It is worth noting that unsupervised representation learning networks, such as an autoencoder (AE) [26], the variants of AE [27]–[29], and a deep belief network (DBN) [30], [31], can learn useful representation for unlabeled data in a low-dimensional space that are of increasing interest to the analysis of high-dimensional HSIs. On account of limited samples for HSIs, unsupervised leaning will become the future trend. AE-based unsupervised learning methods have yielded the state-of-the-art methods for approximating the input, which is the key to target detection in HSIs. Zhao and Zhang [32] exploited stack AE to extract feature matrices of HSIs for anomaly detection. Ozkan *et al.* [33] utilized sparse AE for endmember extraction and hyperspectral unmixing. In particular, variational autoencoders (VAEs) [27] can not only generate an approximated data to the input but also provide a compact data representation to reveal the underlying structure of data. Su *et al.* [34] employed VAE to perform hyperspectral unmixing, which demonstrated very competitive performance compared with other methods. Consequently, a proper design of VAE can help to represent HSIs in a latent space. For the strong representational capability of VAE, we leverage it to reveal the hidden structure of hyperspectral data in deep latent space.

In this article, we propose a deep latent spectral representation learning (DLSRL) based on VAE for band selection to well preserve the physical significance from the original hyperspectral data. It is, to the best our knowledge, the first time to develop and apply VAE for band selection with the help of the deep latent representation. Furthermore, we conduct target detection where target with known spectral signature to evaluate the proposed band selection method. The key of band selection for subsequent target detection is to design the specific and corresponding spectral signature of a known target during band selection. To address this problem, we conduct a simple and powerful selection rule to simultaneously constitute the optimal subset closest to the full bands and the corresponding spectral signature of the known target. Besides, we introduce a hierarchical optimization scheme to further improve the detection accuracy and decrease the false alarm rate. Concretely, we construct an effective stopping rule to break up the iterative optimization scheme. Experiments have been conducted on several data sets. Compared with the other band selection methods, the proposed DLSRL method has made more significant contribution to target detection. We also conduct experiments with different target detection methods and make further efforts to illustrate the superior performance of the proposed band selection for target detection framework. In summary, the contributions of this article are threefold.

1) A novel unsupervised network, DLSRL, is proposed to generate spectral representation closest to the original spectral information in deep latent space that focuses on the issue of insufficient samples and spectral information lost in HSI analysis.
2) An adaptive selection rule is constructed based on the deep latent representation to constitute the optimal subset that can preserve physical significance optimally.
3) A hierarchical optimization scheme is introduced to further improve the detection accuracy and decrease the false alarm rate. We show that the proposed method is superior to the existing state-of-the-art methods.

The remainder of this article is organized as follows. Section II introduces the proposed DLSRL band selection framework. Section III presents a hierarchical optimization scheme for target detection. Section IV presents the experimental results and comprehensive analysis of the proposed method. Section V draws the conclusion.

II. DLSRL Band Selection

Fig. 1 shows the proposed DLSRL network to band selection for hyperspectral data. Let $\mathbf{Y} \in \mathbb{R}^{M\times N\times L}$ and $\hat{\mathbf{Y}} \in \mathbb{R}^{M\times N\times L}$, respectively, represent the normalized and reconstructed HSIs produced by the proposed decoder, where $M\times N$ represents the number of spectral vectors and L is the total number of bands in HSI. Note that all the values in $\mathbf{Y}$ are normalized in the range of (0,1). The deep latent feature matrix is represented by $\mathbf{Z} \in \mathbb{R}^{M\times N\times k}$, where k denotes the number of hidden nodes in the last encoder layer. Let $\mathbf{B} \in \mathbb{R}^{M\times N\times l}$ denote the selected subset bands, where l is the number of selected bands.

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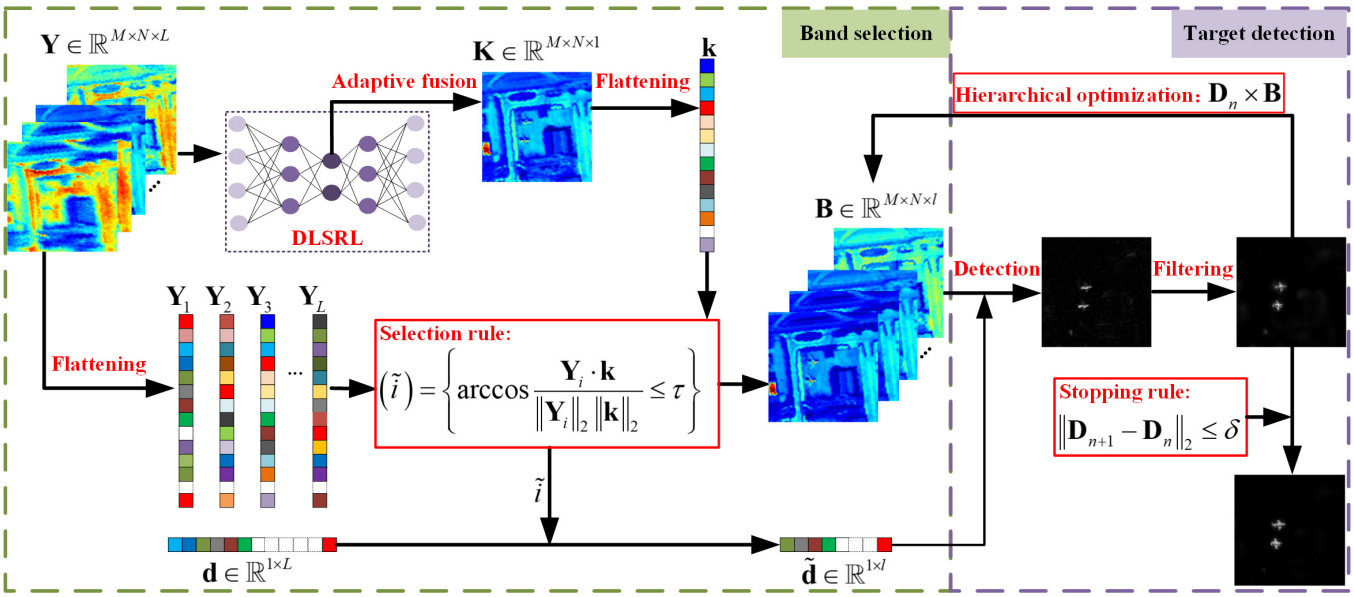


Fig. 1. Overall flowchart of the proposed approach.

A. DLSRL

To discover the hidden feature matrix $\mathbf{Z}$ of an HSI $\mathbf{Y}$ avoiding the shortage of artificially designed image features effectively, $\mathbf{Y}$ is provided into the proposed DLSRL, as shown in Fig. 1, since the VAE can extract the hidden features followed Gaussian distribution in an unsupervised way, which is more in line with hyperspectral characteristics [35]. Although the VAE can map the relationship between the same input and output, it lacks consistency in spectrum that is the key component for HSIs. To build a specific model for the HSIs, a better architecture should be proposed. In this sense, the spectral angle distance is a promising method, because it can map the input spectral vector and its corresponding output one efficiently. Thus, the loss function of the proposed DLSRL can be demonstrated as follows:

$$\begin{aligned}\ell = &-D_{\mathrm{KL}}(q_\phi(\mathbf{Z}|\mathbf{Y})\|p_\theta(\mathbf{Z})) \\ &+\frac{1}{C}\sum_{c=1}^{C}(\log p_\theta(\mathbf{Y}|\mathbf{Z}^c)) \\ &+\frac{1}{\pi}\arccos\frac{\mathbf{Y}\cdot\hat{\mathbf{Y}}}{\|\mathbf{Y}\|_2\|\hat{\mathbf{Y}}\|_2}\end{aligned} \tag{1}$$

where $q_\phi(\mathbf{Z}|\mathbf{Y})$ and $p_\theta(\mathbf{Y}|\mathbf{Z})$ are the probabilistic encoder and decoder networks, respectively. The first term on the left-hand side of (1) is the KL-divergence regularization term, the second one is the reconstruction error in the traditional AE model, and the third one is the physical constraint named spectral representation learning (SRL). $\mathbf{Y}$ is the normalized HSI. It is apparent that the SRL constraint is the average spectral angle distance between $\mathbf{Y}$ and $\hat{\mathbf{Y}}$, which can penalize spectral distance between $\mathbf{Y}$ and $\hat{\mathbf{Y}}$. The lower the value of the spectral angle distance is, the more similar the input and reconstruction ones are. Note that the spectral angle distance is divided by the circular constant π to map the range of the angle from 0 to 1. By introducing the SRL constraint, the deep latent spectral matrix can ensure spectral consistency to a large extent. Fig. 2 shows the curve of our loss function when the epoch changes from 1 to 50. It can be seen that the value of the loss function is between 0 and 1 for the normalized data.

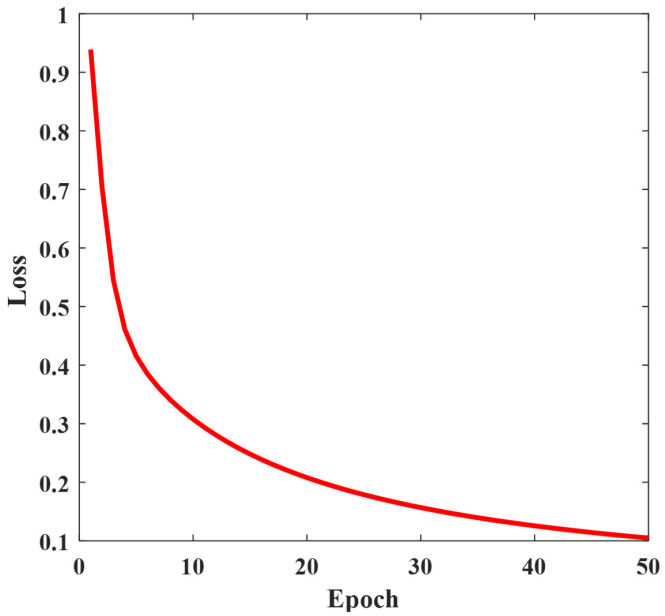


Fig. 2. Curve of the loss function when the epoch changes from 1 to 50.

Considering that $p_\theta(\mathbf{Z})$ and $q_\phi(\mathbf{Z}|\mathbf{Y})$ follow Gaussian distribution, the resulting loss function can also be calculated as:

$$\begin{aligned}\ell \simeq &\frac{1}{2}\sum_{j=1}^{J}(1+\log((\sigma_j)^2)-(\mu_j)^2-(\sigma_j)^2) \\ &+\frac{1}{C}\sum_{c=1}^{C}(\log p_\theta(\mathbf{Y}|\mu_{\hat{\mathbf{Y}}^c},\sigma_{\hat{\mathbf{Y}}^c})) \\ &+\frac{1}{\pi}\arccos\frac{\mathbf{Y}\cdot\hat{\mathbf{Y}}}{\|\mathbf{Y}\|_2\|\hat{\mathbf{Y}}\|_2}\end{aligned} \tag{2}$$

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where $\mathbf{Z}^c = \mu + \sigma \odot \varepsilon^c$ and $\varepsilon^c \sim \mathrm{N}(0, \mathbf{I})$. Here, the symbol "$\odot$" means elementwise product and $\hat{\mathbf{Y}}$ represents the reconstructed version of the input HSI $\mathbf{Y}$. In a VAE model, the input spectral vector passes through the encoder and generates the mean μ and variance σ of the $\mathbf{Y}$ distribution, and the latent feature vector $\mathbf{Z}^c$ is sampled based on the formula $\mathbf{Z}^c = \mu + \sigma \odot \varepsilon^c$. Then, $\mathbf{Z}^c$ is passed into the decoder to reconstruct output $\hat{\mathbf{Y}}$ the same as $\mathbf{Y}$ distribution.

B. Band Selection

By the proposed DLSRL model, the hidden feature matrix can be obtained on the internal layer in an unsupervised way that can fully represent the original HSIs. However, how to make the best use of this latent representation to guide band selection and how to construct an adaptive selection rule are the challenges. To address these problems, we propose a novel framework. First, the latent feature matrix $\mathbf{Z}$ is adaptively fused to obtain a comprehensive feature map $\mathbf{K}$. Considering that each hidden node provides different information contents, we adopt an adaptive fusion method based on structure tensor (ST), which was first proposed in our previous work [36] as

$$\mathbf{K} = \sum_{i=1}^{k} \omega_i \mathbf{Z}_i \tag{3}$$

where the weight coefficient ω_i is specifically defined by the larger eigenvalue of the ST matrix in the feature map $\mathbf{Z}_i$, that is

$$\omega_i = \frac{\beta_i}{\sum_i^k \beta_i} \tag{4}$$

where β_i represents the larger eigenvalue of the ST matrix for $\mathbf{Z}_i$ that can be decomposed as

$$\mathbf{S}^i = \begin{bmatrix} \eta_i & \upsilon_i \end{bmatrix} \begin{bmatrix} \beta_i & 0 \\ 0 & \lambda_i \end{bmatrix} \begin{bmatrix} \eta_i \\ \upsilon_i \end{bmatrix} \tag{5}$$

where β_i and λ_i are the nonnegative eigenvalues of $\mathbf{Z}_i$, and η_i and υ_i are their corresponding eigenvectors. β_i is the larger eigenvalue, which illustrates the response intensity of $\mathbf{Z}_i$, and the corresponding eigenvector, η_i, indicates the orientation that maximizes the spectral value fluctuations. If $\beta_i \geq \beta_{i+1}$, the response intensity of the ith feature map is larger. Therefore, it should occupy a larger proportion of weighting. Conversely, the $(i+1)$th feature map should occupy a larger proportion of weighting in the case of $\beta_i < \beta_{i+1}$. More details are described in our previous work [36]. The comprehensive feature map $\mathbf{K}$ and each band of the original HSI $\mathbf{Y}$ are both flattened to vectors. Then, we construct an adaptive selection rule to obtain the optimal sequence number of band subset as

$$(\tilde{i}) = \left\{ \arccos \frac{\mathbf{Y}_i \cdot \mathbf{k}}{\|\mathbf{Y}_i\|_2 \|\mathbf{k}\|_2} \leq \tau \right\} \tag{6}$$

where $\tilde{i}$ represents the sequence number of the selected band, $\mathbf{Y}_i$ is the flattened ith band of the original HSI $\mathbf{Y}$, and $\mathbf{k}$ denotes the flattened feature map. Here, τ is a threshold. When the spectral angle distance between $\mathbf{Y}_i$ and $\mathbf{k}$ is less than or equal to τ, the band $\mathbf{Y}_i$ is selected and the sequence number is defined as $\tilde{i}$. All bands from the original HSI $\mathbf{Y}$ satisfying the above formulas are set to the selected optimal subset $\mathbf{B} \in \mathbb{R}^{M \times N \times l}$, where l is the number of the selected bands. Note that a threshold is set to select band subset by the minimum spectral angle distance between $\mathbf{Y}_i$ and $\mathbf{k}$. Since the proposed DLSRL can fully represent the original HSIs by addressing the physical constraint, the resulting subset $\mathbf{B}$ can not only preserve physical significance but also represent full bands' information optimally.

III. Target Detection

In accordance with the above procedure, the mapping relationship between the deep latent representation and the optimal band subset is effectively built for the first time. It is also of interest to evaluate the influence of the DLSRL-based band selection method on target detection. Many algorithms have been developed for target detection in HSI, such as adaptive coherence estimator (ACE) [37], orthogonal subspace projection (OSP) [38], constrained energy minimization (CEM) [39], hierarchical CEM (hCEM) [40], and a combined sparse and collaborative representation (CSCR) [41]. In this section, we introduce a hierarchical optimization framework to further improve the detection accuracy and decrease the false alarm rate, as shown in Fig. 1. With respect to target detection, a known target spectrum $\mathbf{d} \in \mathbb{R}^{1 \times L}$ is given. To comply with the selected band number $\tilde{i}$, the $\tilde{i}$th pixel in the spectral domain of $\mathbf{d}$ is also selected as $\tilde{\mathbf{d}} \in \mathbb{R}^{1 \times l}$. As shown in Fig. 1, each detection procedure consists of two steps, i.e., detection and filtering. Here, a typical algorithm, CEM, is employed to obtain an initial detection map

$$\mathbf{D}_{\mathrm{ini}} = \frac{\tilde{\mathbf{d}}^T \mathbf{R}^{-1}}{\tilde{\mathbf{d}} \mathbf{R}^{-1} \tilde{\mathbf{d}}^T} \mathbf{B}. \tag{7}$$

CEM designs a finite impulse response (FIR) filter $\mathbf{R}$ to pass through the selected subset while minimizing its output energy, except for the desired target $\tilde{\mathbf{d}}$. Considering that adjacent pixels are highly correlated, a guided filter (GF) [42] is conducted on $\mathbf{D}_{\mathrm{ini}}$ to avert artifacts at the boundaries and consider the neighboring spatial correlations, that is

$$\mathbf{D}_1 = \mathrm{GF}(\mathbf{D}_{\mathrm{ini}}, \mathbf{D}_{\mathrm{ini}}). \tag{8}$$

The guidance image is defined as $\mathbf{D}_{\mathrm{ini}}$ itself. Here, we set the filer size and the blur degree as 7 and 0.8^2, respectively. $\mathbf{D}_1$ contains more target information rather than background, which can suppress the background. Therefore, we use it as a weight matrix to suppress the background. In particular, $\mathbf{D}_1$ is first multiplied by the selected subset of bands $\mathbf{B}$. The multiplied result is input to the CEM detector. Then, the GF is adopted to get the second detection result $\mathbf{D}_2$. The similarity between the two detection results $\mathbf{D}_1$ and $\mathbf{D}_2$ is compared, and if they are similar, the detection is stopped, and if not, the above process is repeated. This hierarchical optimization

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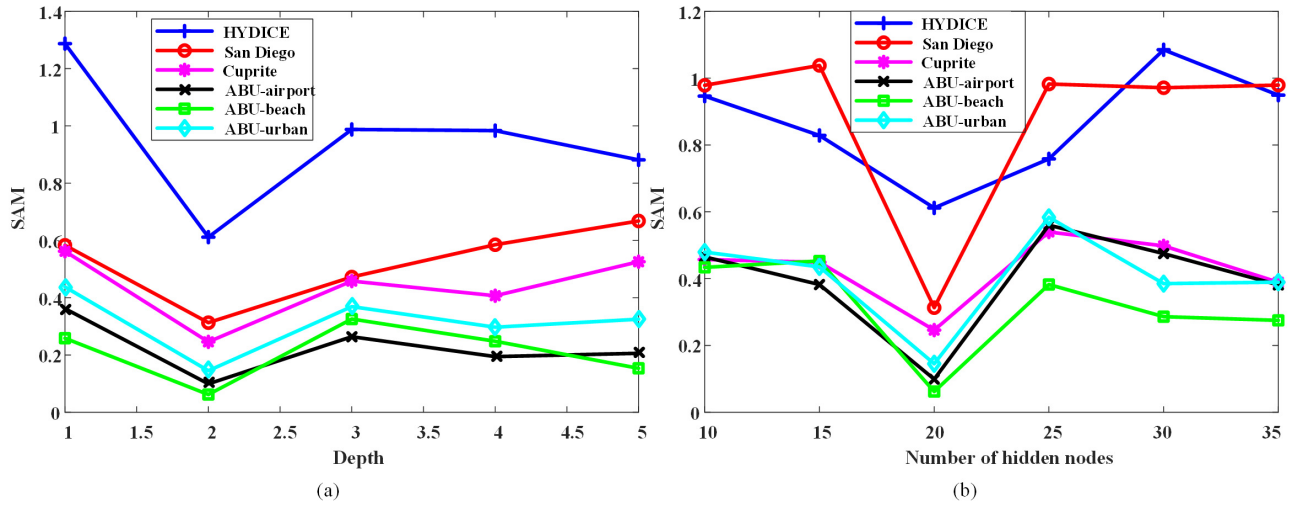


Fig. 3. Parameter sensitivity analysis for the parameters (a) depth and (b) number of hidden nodes, respectively.

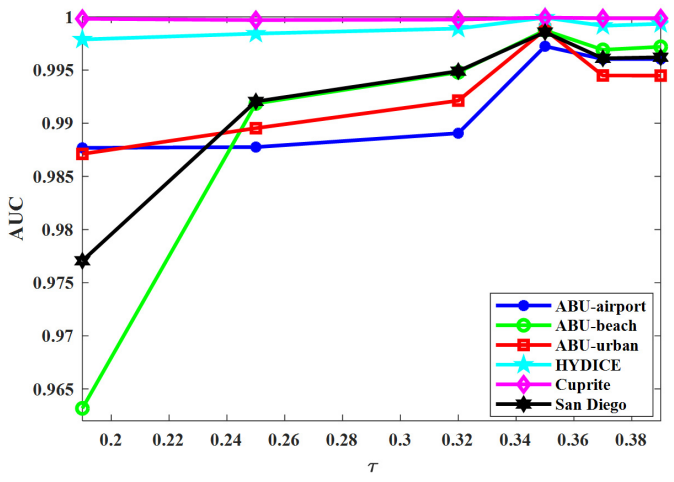


Fig. 4. Effect of τ on target detection.

for target detection aims to further reduce the false alarm rate. The stopping rule is defined as

$$\|\mathbf{D}_{n+1} - \mathbf{D}_n\|_2 \leq \delta \tag{9}$$

where $\mathbf{D}_{n+1}$ and $\mathbf{D}_n$ indicate the $n+1$th and nth detection maps, respectively. The experiments show that the value of n is generally 4 or 5. Here, δ is a small positive number, which is set to 10^{-4}. When the current detection result is sufficiently similar to the previous detection result, the hierarchical optimization process will stop.

TABLE I

AUC VALUES OF (P_D, P_F) FOR THE COMPARED BAND SELECTION METHODS ON TARGET DETECTION

Images	Proposed	E-FPDC	LCMV-BCC	MVPCA	OCF
ABU-a1	**0.99231**	0.98344	0.98229	0.92985	0.97044
ABU-a2	**0.99945**	0.99101	0.98885	0.98858	0.99899
ABU-a3	**0.99724**	0.99369	0.98428	0.97927	0.98855
ABU-a4	**0.99998**	0.99731	0.99641	0.99102	0.99653
ABU-b1	**0.99996**	0.99358	0.98892	0.98929	0.98993
ABU-b2	**0.99771**	0.99028	0.99097	0.98790	0.98353
ABU-b3	**1**	0.99889	0.99762	0.99773	0.99563
ABU-b4	**0.99728**	0.98592	0.98615	0.98615	0.98737
ABU-u1	**0.99951**	0.99544	0.99664	0.99669	0.99612
ABU-u2	**0.99957**	0.99341	0.99244	0.99346	0.99325
ABU-u3	**0.99996**	0.99816	0.99788	0.99435	0.99797
ABU-u4	**0.99896**	0.98965	0.98857	0.98958	0.99003
ABU-u5	**0.99602**	0.97617	0.96701	0.97663	0.97909
HYDICE	**0.99993**	0.98434	0.98729	0.95718	0.98422
San Diego	**0.99984**	0.98614	0.98930	0.98374	0.98594
Cuprite	**0.99994**	0.98939	0.98471	0.85552	0.98592

TABLE II

AVERAGE RUNNING TIME (S) FOR THE COMPARED BAND SELECTION METHODS ON TARGET DETECTION

Images	Proposed	E-FPDC	LCMV-BCC	MVPCA	OCF
Average	**0.61150**	0.89238	2.24413	0.71581	25.7376

IV. EXPERIMENTAL RESULTS

A. Data Sets

Four real data sets acquired by different sensors over different scenes are utilized, which are listed as follows.

1) San Diego Data Set: The San Diego data set is widely used to evaluate target detection performance, which is collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the San Diego airport area, CA, USA. This data set collects 224 spectral channels spanning from 400 to 2500 nm, and after removing the water vapor absorption and noisy bands, 189 bands are remained. There are 100 × 100 pixels in each band. All the compared algorithms are used to detect three airplanes.

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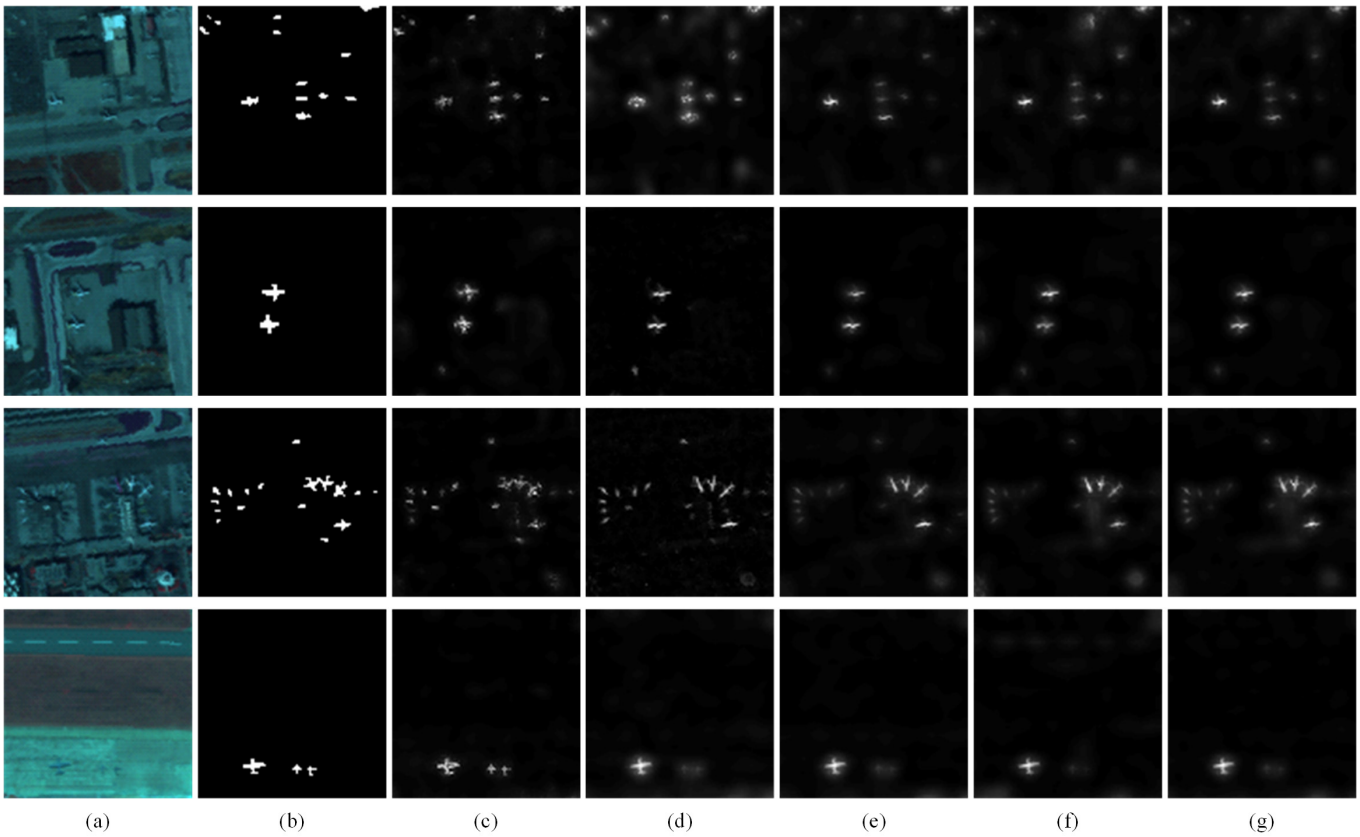


Fig. 5. Detection maps of different band selection methods for target detection. (a) Color composites of ABU-airport scenes. (b) Reference map. (c) Proposed. (d) E-EPDC. (e) LCMV-BCC. (f) MVPCA. (g) OCF.

2) Cuprite Data Set: The Cuprite[1] data set was also acquired by the AVIRIS sensor in the Cuprite mining district of Nevada in 1997. Here, a subset of Cuprite data containing 250 × 191 pixels and 188 bands (low SNR and water absorption bands were removed) is utilized in the experiments. The buddingtonite is considered as the target that should be detected.

3) Airport-Beach-Urban (ABU) Database: The ABU database was made available online[2]. This database consists of 13 HSIs with details described in [43]. These 13 sample images contain 100 × 100 or 150 × 150 pixels in the spatial domain with the corresponding references. The airplane and urban scenes were captured by the AVIRIS sensor, and most of the beach scenes are also captured by the AVIRIS sensor. One of the beach scenes was captured by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor.

4) HYDICE Data Set: This data set was acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor over an urban area, CA, USA. A subimage of size 80 × 100 is widely used in many publications[3]. This data set has 210 bands, but only 162 channels were left after removing the noisy bands. Among them, 21 pixels are considered as targets.

[1]http://aviris.jpl.nasa.gov/html/aviris.freedata.html

[2]http://xudongkang.weebly.com/

[3]http://www.erdc.usace.army.mil/Media/FactSheets/FactSheetsArticleView/tabid/9254/Article/476681/hypercube.

B. Parameter Settings

For the proposed DLSRL model, the depth and the number of hidden nodes determine the quality of the learned features in the spectral domain. Consequently, we calculate spectral angle mapper (SAM) [44] between the input spectral vector and the reconstructed spectral vector to set these two parameters. The ideal value of SAM is 0. Fig. 3 shows the effect of the depth and the number of hidden nodes on the SAM, when one of the parameters is analyzed and the value of the other parameter remains stable. It can be observed that when the depth reaches 2 and the number of hidden nodes is 20, our proposed DLSRL model can achieve best reconstruction performance, which means that the deep latent representation in this case can best characterize the spectral information of the original HSIs. Therefore, the depth is set as 2 and the number of hidden nodes is fixed as 20 for all data sets.

Besides, the threshold τ in the constructed selection rule determines the number of the selected bands. Since we evaluate the influence of the DLSRL-based band selection method on target detection, the receiver operating characteristic (ROC) and the area under the curve (AUC) metric [45], [46] are used. The AUC value of ROC curve of (P_D, P_F) is utilized to set τ.

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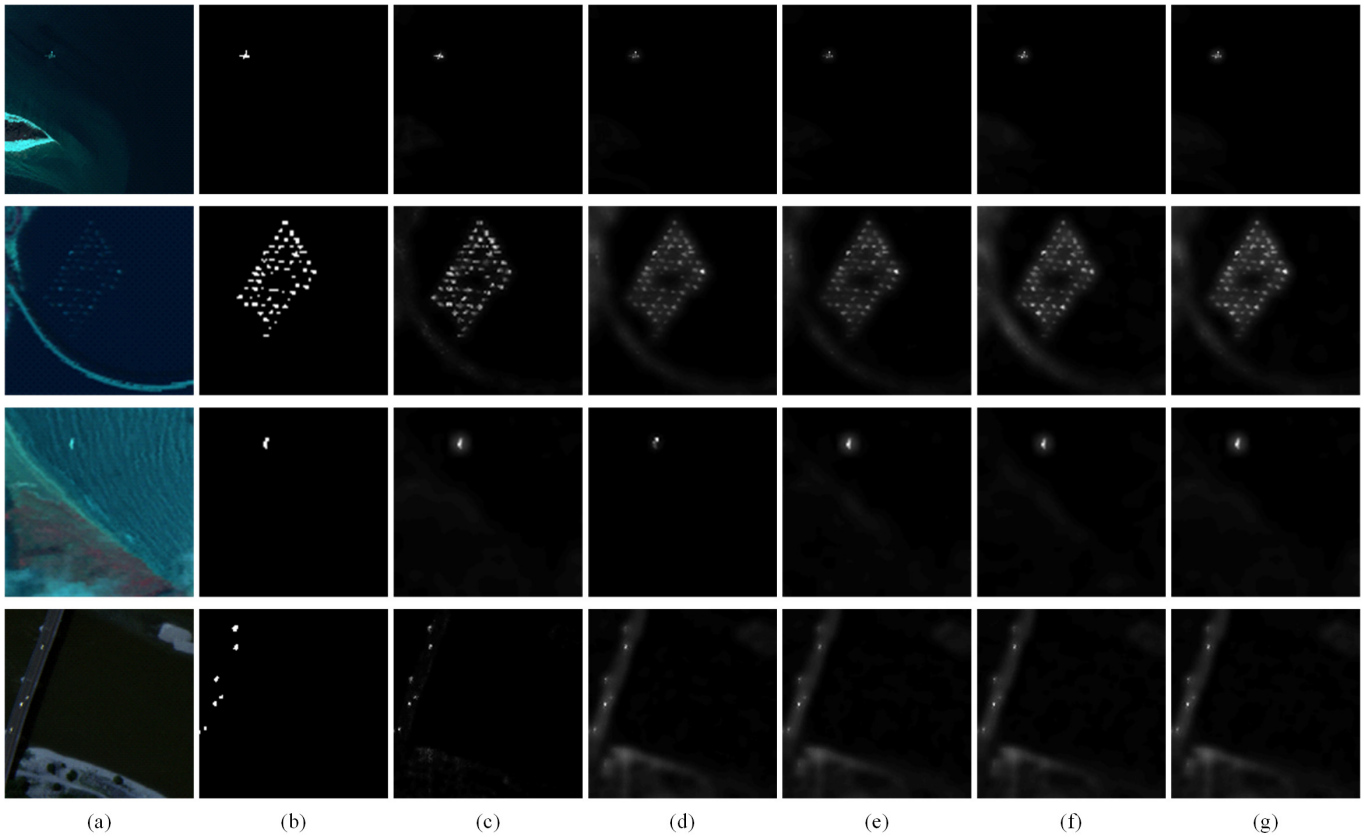


Fig. 6. Detection maps of different band selection methods for target detection. (a) Color composites of ABU-beach scenes. (b) Reference map. (c) Proposed. (d) E-EPDC. (e) LCMV-BCC. (f) MVPCA. (g) OCF.

Fig. 4 shows the change in the AUC value with different values of τ for all data sets. It can be observed that the best AUC value is achieved when τ is 0.35 for all data sets. Hence, we set τ as 0.35.

C. Detection Results

The ROC curve describes detection performance with the true positive rate (P_D) and the false positive rate (P_F) at each threshold (t). The (P_D,P_F) curve describes the tradeoff between P_D and P_F. The (P_F,t) curve indicates the false alarm rate of the target detection. The larger the value of the AUC (P_D,P_F), the better the detection performance. The smaller the AUC (P_F,t), the lower the probability of false detection. These indexes are used to evaluate the detection performance.

1) Different Band Selection Methods: To evaluate the performance of the proposed band selection method, comprehensive and comparative experiments are conducted on four representative unsupervised band selection methods: E-EPDC, LCMV-BCC, MVPCA, and OCF. These different band selection methods are conducted on our target detection method. Implementation of these methods was achieved via the publicly released codes. The quantitative comparison results of all the considered algorithms are listed in Table I. It can be observed that the proposed band selection method is more suitable for target detection, which can achieve highest AUC values of (P_D,P_F) for all data sets. It also shows that the proposed method has best generalization. In addition, two operations, including band selection and target detection, are used to calculate the average time on all the testing images reported in Table II. When the target detection method remains unchanged, the proposed band selection method is the fastest in comparison to the E-FPDC, LCMV-BCC, MVPCA, and OCF methods. Note that the bolder ones mean the best.

The reference map and detection maps are shown in Figs. 5–8. From these figures, it can be clearly seen that our proposed method causes the smallest visual difference between the reference and detection map in comparison with E-EPDC, LCMV-BCC, MVPCA, and OCF methods. More importantly, the proposed method can discover the targets of different structures. For instance, the point target in the HYDICE data set and the targets with area information in other data sets can all be effectively detected. Besides, the detection maps of the proposed method contain minimum background reaching the lowest false alarm rate for all the testing HSIs.

2) Different Target Detection Methods: To further verify the effectiveness of the proposed target detection method, four representative target detection methods, including CEM, hCEM, CSCR, and FFSM, are utilized for comparison with the selected subset. These competing methods are either frequently cited in the literature or have leading performances on several real hyperspectral data sets. Tables III and IV

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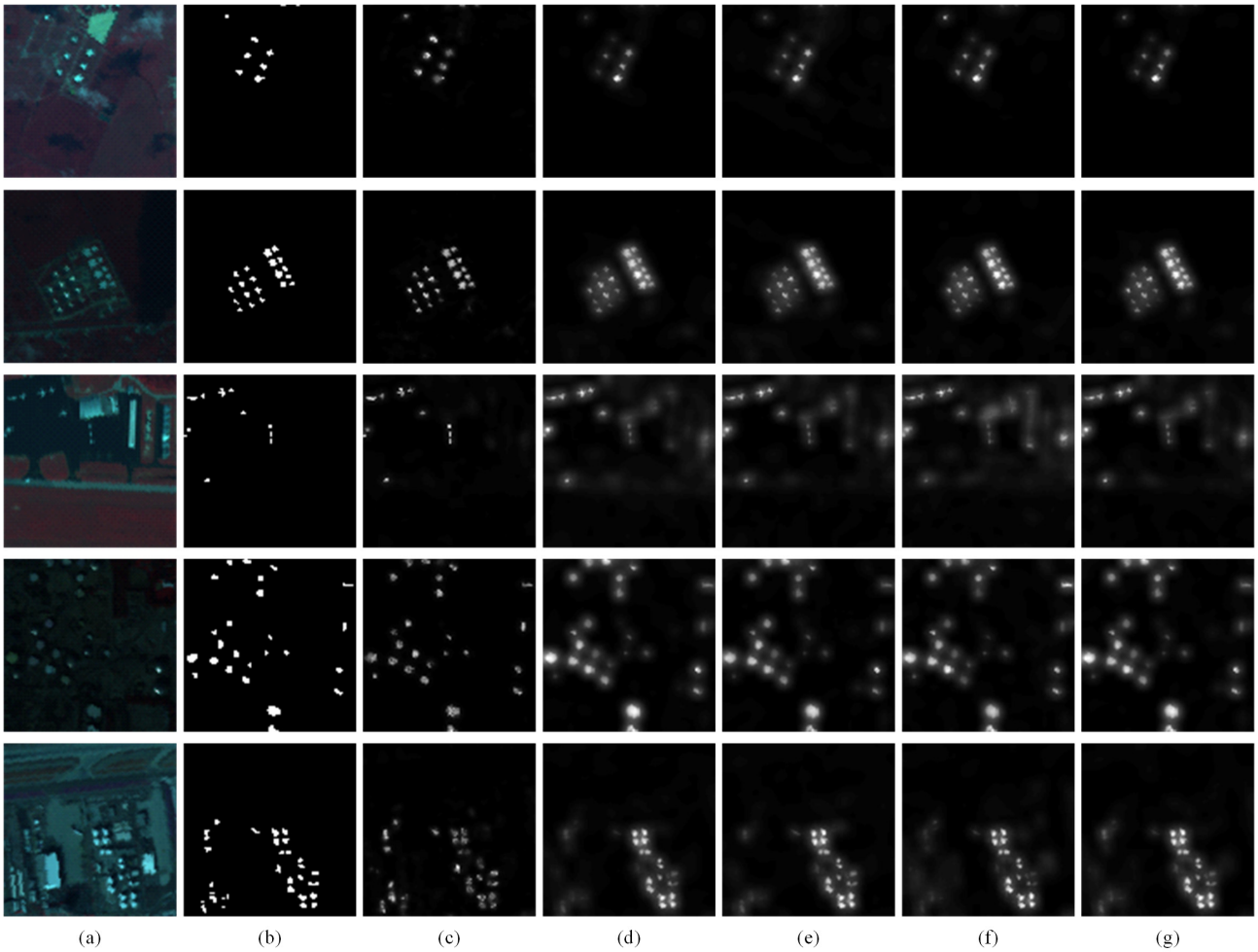


Fig. 7. Detection maps of different band selection methods for target detection. (a) Color composites of ABU-urban scenes. (b) Reference map. (c) Proposed. (d) E-EPDC. (e) LCMV-BCC. (f) MVPCA. (g) OCF.

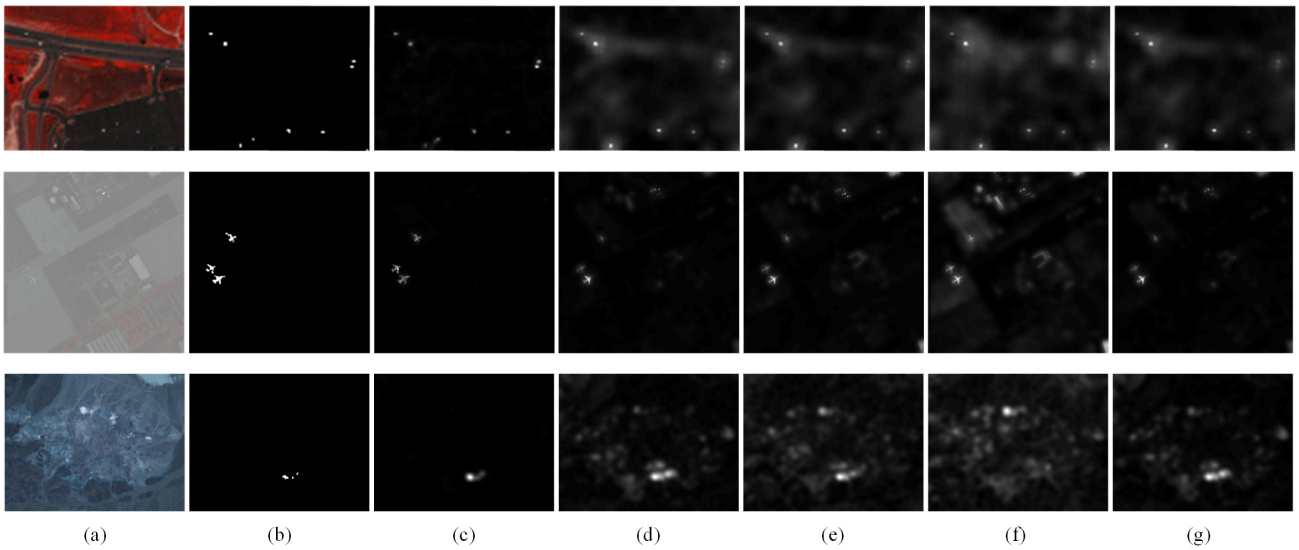


Fig. 8. Detection maps of different band selection methods for target detection. (a) Color composites of HYDICE, San Diego, and Cuprite data sets. (b) Reference map. (c) Proposed. (d) E-EPDC. (e) LCMV-BCC. (f) MVPCA. (g) OCF.

report the AUC values of (P_D,P_F) and (P_F,t) obtained by the comparative detection methods, where best values are boldfaced. From Table III, the AUC value of (P_D,P_F) obtained by the proposed method is also the highest for all the scenes compared with those obtained by BS-CEM, BS-hCEM, BS-CSCR, and BS-FFSM. Here, BS represents the

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TABLE III
AUC VALUES OF (P_D,P_F) FOR THE COMPARED TARGET DETECTION METHODS WITH THE SAME SUBSET

Images	Proposed	BS-CEM	BS-hCEM	BS-CSCR	BS-FFSM
ABU-a1	**0.99231**	0.95695	0.89504	0.86645	0.84108
ABU-a2	**0.99945**	0.98310	0.98352	0.83075	0.84276
ABU-a3	**0.99724**	0.98807	0.94610	0.86305	0.90312
ABU-a4	**0.99998**	0.99884	0.92662	0.88068	0.94556
ABU-b1	**0.99996**	0.97747	0.99080	0.99545	0.96832
ABU-b2	**0.99771**	0.99520	0.98073	0.97426	0.95373
ABU-b3	**1**	**1**	0.99778	0.99926	0.82034
ABU-b4	**0.99728**	0.95846	0.93893	0.73381	0.81322
ABU-u1	**0.99951**	0.99933	0.96480	0.99075	0.94819
ABU-u2	**0.99957**	0.99941	0.99072	0.76546	0.98816
ABU-u3	**0.99996**	0.99948	0.98580	0.90669	0.94745
ABU-u4	**0.99896**	0.99843	0.99622	0.96271	0.98763
ABU-u5	**0.99602**	0.99402	0.97658	0.95895	0.91381
HYDICE	**0.99993**	0.99984	0.99873	0.98424	0.95621
San Diego	**0.99984**	0.97718	0.98874	0.98502	0.95044
Cuprite	**0.99994**	**0.99994**	0.99941	0.97861	0.99564

TABLE IV
AUC VALUES OF (P_F,t) FOR THE COMPARED TARGET DETECTION METHODS WITH THE SAME SUBSET

Images	Proposed	BS-CEM	BS-hCEM	BS-CSCR	BS-FFSM
ABU-a1	**0.01083**	0.03049	0.03259	0.03329	0.07581
ABU-a2	**0.01097**	0.01864	0.02512	0.04985	0.05128
ABU-a3	**0.00923**	0.02587	0.02934	0.01331	0.01379
ABU-a4	**0.00317**	0.02352	0.03898	0.15395	0.01091
ABU-b1	**0.00471**	0.01572	0.02782	0.03029	0.01879
ABU-b2	**0.00642**	0.02478	0.02412	0.05451	0.02159
ABU-b3	**0.00446**	0.01038	0.01374	0.01285	0.05691
ABU-b4	**0.01099**	0.03632	0.03721	0.03473	0.05272
ABU-u1	**0.00658**	0.01054	0.02354	0.04365	0.08001
ABU-u2	**0.00534**	0.01654	0.02783	0.01761	0.04449
ABU-u3	**0.01029**	0.02371	0.02831	0.03105	0.07165
ABU-u4	**0.01018**	0.02527	0.02765	0.01594	0.01503
ABU-u5	**0.01091**	0.02752	0.02988	0.03723	0.09157
HYDICE	**0.00839**	0.02135	0.02542	0.03932	0.06232
San Diego	**0.00667**	0.02023	0.02132	0.01991	0.03321
Cuprite	**0.01587**	0.04212	0.04531	0.19454	0.17744

selected subset by our proposed band selection method. The AUC value of (P_F,t) can evaluate the effect of background suppression and reflect the false alarm rate. It can be observed from Table IV that the proposed method achieves smaller AUC values of (P_F,t) for all data sets, which illustrates that the proposed band selection method can decrease the false alarm rate. Besides, Table V lists the average running time of different target detection methods. In comparison to CEM, the proposed target detection method is more time-consuming, but illustrates the improvement in detection accuracy and false alarm rate. In addition, the proposed target detection method is faster than the hCEM, CSCR, and FFSM methods.

TABLE V
AVERAGE RUNNING TIME (S) FOR THE COMPARED TARGET DETECTION METHODS WITH THE SAME SUBSET

Images	Proposed	CEM	hCEM	CSCR	FFSM
Average	0.61150	**0.60344**	9.87231	17.0546	79.3080

With the optimal parameters, Fig. 9 shows the ROC of (P_D,P_F) obtained by different detectors using two typical and widely used data sets, i.e., HYDICE and San Diego. These two data sets are representative of targets at the subpixel level and targets with full-size pixels, respectively. From the results, it can be clearly seen that the proposed DLSRL-based detection method is always superior to the comparative methods. This result confirms that our method has promising results for these scenes, which also illustrates that the proposed detector based on DLSRL has better generalization ability. The ROC curve of (P_F,t) represents the false alarm rate, as shown in Fig. 10. It can be observed that the proposed method achieves smaller AUC values of (P_F,t), which demonstrates that the proposed method can suppress background better with best detection results.

D. Component Analysis

This section analyzes the effects of the major processing components over the AUCs on each data set. There are six comparative experiments. The first is to input the full bands of HSIs into hCEM. The second is to input the full bands into CEM. These two experiments can illustrate which detector performs better, and we choose the detector with a better performance as the initial detection procedure. Furthermore, CEM is easier than hCEM. The third and fourth are the combination of BS methods and the CEM detector. In particular, the third is the band selection method without spectral constraint loss, i.e., BS-CEM, and the fourth is with spectral constraint loss, i.e., BS_S-CEM. These two experiments can evaluate the influence of spectral constraint loss on the final result. The fifth is the proposed hierarchical optimization scheme with a Gaussian filter (i.e., Proposed_GS) and the sixth with a GF (i.e., Proposed_GF) is to show that the GF is better than the Gaussian filter in the filtering step. The AUC values of (P_D,P_F) and (P_F,t) are listed in Tables VI and VII, respectively. The better the detection, the higher the AUC values of (P_D,P_F), and the smaller the AUC value of (P_F,t). The ideal AUC value of (P_D,P_F) is 1 and the ideal AUC value of (P_F,t) is 0. It can be observed from Table VI that each component has positive effect on target detection for all data sets. In addition, the false alarm rate is also decreased by the proposed framework, as listed in Table VII.

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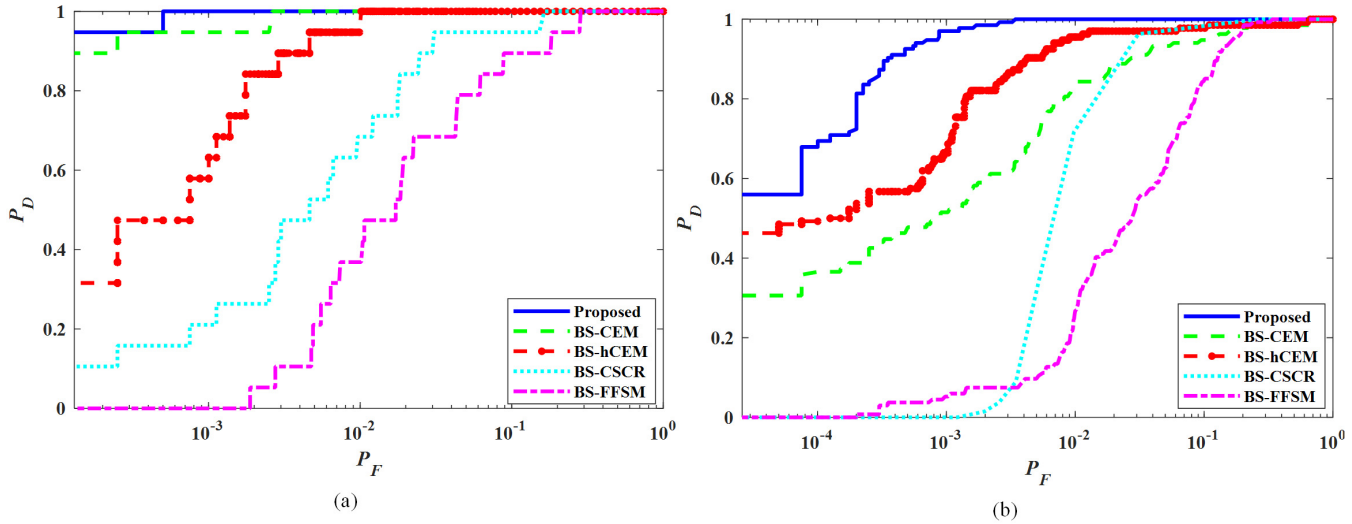


Fig. 9. ROC curves of (P_D, P_F) obtained by the compared methods on (a) HYDICE and (b) San Diego data sets.

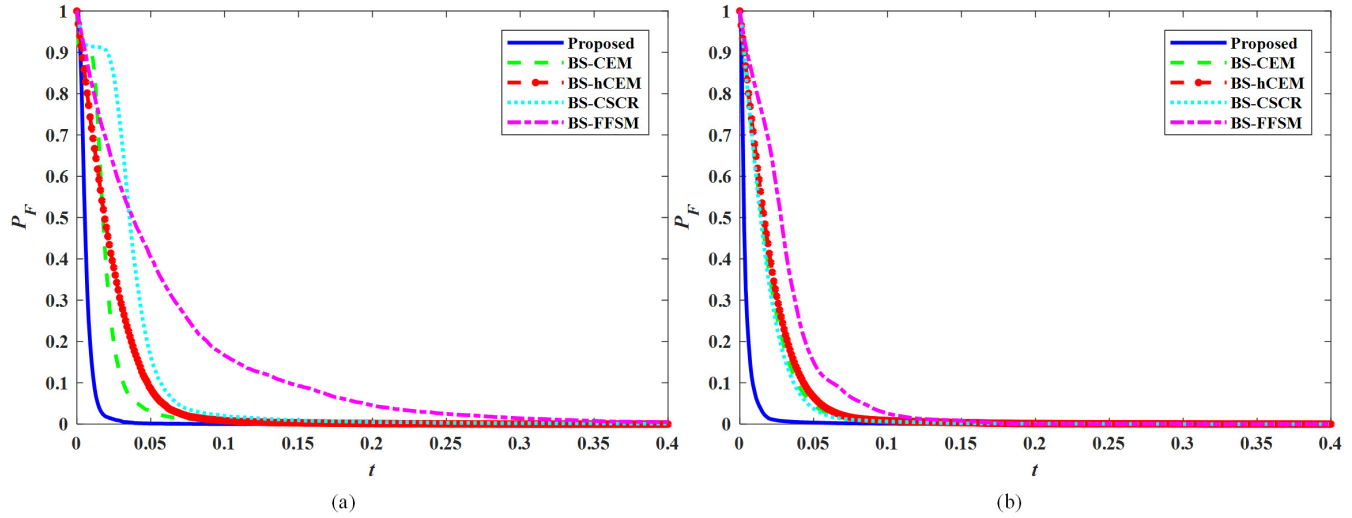


Fig. 10. ROC curves of (P_F, t) obtained by the compared methods on (a) HYDICE and (b) San Diego data sets.

TABLE VI
AUC VALUES OF (P_D, P_F) FOR THE COMPONENT ANALYSIS

Methods	ABU	HYDICE	San Diego	Cuprite	Average
hCEM	0.96579	0.99918	0.99119	0.99939	0.98889
CEM	0.98755	0.99984	0.97718	0.99994	0.99113
BS-CEM	0.98804	0.99986	0.98691	0.99994	0.99369
BS_S-CEM	0.99009	0.99987	0.98779	0.99994	0.99442
Proposed_GS	0.99557	0.99982	0.99857	0.99994	0.99848
Proposed_GF	0.99826	0.99993	0.99984	0.99994	0.99950

TABLE VII
AUC VALUES OF (P_F, t) FOR THE COMPONENT ANALYSIS

Methods	ABU	HYDICE	San Diego	Cuprite	Average
hCEM	0.03024	0.02685	0.02492	0.04682	0.03221
CEM	0.02551	0.02462	0.02218	0.04381	0.02903
BS-CEM	0.02427	0.02354	0.02127	0.04353	0.02815
BS_S-CEM	0.02238	0.02135	0.02023	0.04212	0.02652
Proposed_GS	0.01237	0.01364	0.01662	0.02492	0.01689
Proposed_GF	0.00795	0.00839	0.00667	0.01587	0.00972

The results reported in Tables VI and VII demonstrate that each component is significant for target detection and the proposed method can decrease false alarm rate with the best detection results.

V. CONCLUSION

In this article, we present a novel band selection method called DLSRL, which is leveraged to search the closest representation of the original HSIs by imposing multivari-

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ate Gaussian and spectral consistency on deep latent space. An adaptive selection rule is proposed to construct the mapping relationship between the deep latent representation and the optimal subset that can preserve physical significance optimally. A hierarchical optimization technique is conducted on the selected subset to detect targets. The experiments on four real data sets show that the proposed band selection method has powerful influence on target detection. Further comparative experiments validate that the proposed method is effective for detecting targets presented at the pixel level as well as those with structural information. In the future, we will analyze the sensitive bands for different detected objects.

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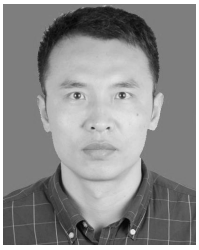
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